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An NLP-Based Method for Assessment of GDPR Compliance in Data Privacy Agreements

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Abstract

In the digital era, the massive utilization of personal data has raised significant privacy and compliance concerns, underscored by frequent data breaches and privacy violations. The introduction of the General Data Protection Regulation (GDPR) by the European Union marks a stringent global standard for data protection, emphasizing the need for rigorous compliance mechanisms. Traditional methods for assessing GDPR compliance within Data Processing Agreements (DPAs) are often manual, time-consuming, and prone to human error. This paper introduces a novel NLP-based methodology that leverages the BERT text classification model to automate the assessment of GDPR compliance in DPAs. Our method not only expedites the process but also enhances accuracy and reduces the potential for oversight. Empirical results demonstrate the method's efficacy, with high precision and recall in detecting compliance issues, indicating a significant step forward in automating legal document analysis. This advancement holds substantial promise for streamlining GDPR compliance checks, providing legal professionals and data protection officers with a powerful tool that augments the accuracy and efficiency of their work.

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CCS Concepts

• **Security and privacy** → Security services; Privacy-preserving protocols.

Keywords

GDPR, Data Processing Agreements, Compliance, Natural Language Processing

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1 Introduction

With the rapid development of information technology and the deepening of digital transformation, data has become a key asset in modern economy. Enterprises and organizations rely on big data analysis to insight market trends, predict consumer behavior, improve service quality, and develop personalized products and services. This process inevitably involves the collection, processing, and transmission of a large amount of personal data, including but not limited to personal identity information, location data, consumption habits, online behavior, and even biometric information [1, 2].

However, the widespread use of personal data also brings significant challenges to privacy and security. The frequent occurrence of data leaks, misuse of personal information, and infringement of privacy has attracted great public attention. The security of personal data is not only related to the protection of personal privacy rights, but also involves the right of data subjects to know and control how their data is collected, used, and shared [3]. Ensuring the security and compliance of personal data during collection, processing, and

transmission has become particularly important. In the context of data security, various methodologies have been developed to enhance data collection and processing efficiency. Yu et al. proposed a sensing data collection protocol for smart distribution grids based on the Named Data Networking (NDN) framework [4]. This approach leverages the inherent advantages of NDN to improve data collection efficiency and reliability.

The General Data Protection Regulation (GDPR) is a data protection and privacy law implemented by the European Union (EU) on May 25, 2018, aimed at strengthening and unifying the protection of all personal data within Europe [5]. GDPR applies not only to organizations within Europe, but also to non-European organizations processing EU citizen data, making it one of the strictest data protection regulations worldwide [6].

The core principles of GDPR include data minimization, data accuracy, limitations on processing purposes, data security, and transparency. It requires organizations to take appropriate technical and organizational measures to protect personal data, ensure that data processing activities are both legal and fair, and transparent to data subjects. In addition, GDPR grants data subjects a series of rights, including access rights, correction rights, deletion rights, etc. GDPR imposes extremely severe penalties on non-compliant organizations, forcing them to handle personal data with extreme caution and ensuring that all their data processing activities comply with GDPR requirements.

Under the framework of GDPR, Data Processing Agreement (DPA) has become a key document for collaboration between organizations [7]. The DPA specifies the responsibilities and obligations between data processors and data controllers to ensure that both parties comply with GDPR regulations when processing personal data. The content involved in DPA includes but is not limited to the purpose of data processing, data types, processing duration, rights of data subjects, and security measures for data transfer. Accurately assessing and verifying the GDPR compliance of DPA is one of the key steps in ensuring the legitimate processing of personal data by organizations [8]. However, manually reviewing DPA to verify its compliance with GDPR requirements is both time-consuming and error prone. With the development of technology, especially the advancement of natural language processing (NLP) technology, it has provided possibilities for automated processing and analysis of large amounts of legal texts [9, 10]. The automation method based on NLP can effectively parse DPA documents, identify key terms and information, and evaluate their consistency with GDPR requirements, greatly improving the efficiency and accuracy of compliance checks [11, 12]. This research direction is not only of great significance to legal professionals and data protection officials, but also provides new application scenarios for machine learning and artificial intelligence fields [13, 14].

In this study, we propose an innovative method for GDPR compliance evaluation using BERT based text classification technology to automate the Data Processing Protocol (DPA). BERT is an advanced pre trained deep learning model that has shown excellent performance in various natural language processing tasks. Our method utilizes the powerful semantic understanding ability of the BERT model to parse and classify key information in DPA documents, such as the nature of data processing activities, the rights of data subjects, and security measures. The development of this method not only provides strong tools for legal professionals and data protection officials to ensure more efficient and accurate GDPR compliance assessments, but also opens up new paths for the practical application of machine learning and artificial intelligence in the legal field. This demonstrates the potential and value of artificial intelligence technology in addressing legal compliance issues.

The remaining structure of this paper is as follows. The second chapter introduces the related methods. The third chapter introduces the GDPR compliance evaluation method in DPA. The fourth chapter is an empirical evaluation of compliance inspection methods. Finally, the conclusion is in the fifth chapter.

2 Related Methods

Automated compliance inspection methods have received widespread attention in recent years due to their potential to improve efficiency and accuracy [15, 16]. The automation method aims to reduce the need for manual labor through technological means and improve the speed and accuracy of compliance testing. Xi et al. proposed a method for locational detection of false data injection attacks in power grids using a multi-label cascade boosting forest integrated with relevant features and extreme gradient boosting [17]. This approach enhances the accuracy of detecting complex anomalies, which is analogous to our goal of improving compliance detection accuracy in DPAs through advanced NLP techniques.

Cejas et al. developed an innovative solution leveraging NLP technologies to automate the compliance checking of DPAs against GDPR [18]. Their methodology involves the creation of two foundational artifacts: the extraction of "shall" requirements from GDPR relevant to DPAs, and a glossary defining legal concepts. The core of their approach is a sophisticated algorithm that compares phrasal-level representations of the DPA's content against predefined GDPR "shall" requirements, identifying non-compliance and recommending corrections.

Chhetri et al. discussed the challenges faced by organizations due to GDPR enforcement, particularly in demonstrating compliance and automated compliance verification [19]. And it provides an extensible tool for automatic compliance verification based on informed consent, utilizing knowledge graphs and regulatory to code processes to transform GDPR regulations into technical measures and software code.

Liu et al. introduced a novel approach to automate the structural analysis of privacy policies for Android apps using NLP technologies [20]. Their work involves the development of a manually labeled corpus containing 167 privacy policies with over 447K words and 5,276 annotated paragraphs. The foundation of their methodology is a classification scheme based on GDPR articles, which characterizes the topics for paragraphs within these policies. They employed four document classification models, including a hierarchical model that integrates BERT for sentence representation and BiLSTM for paragraph representation. This sophisticated system automatically categorizes paragraph topics, aiding in the understanding and restructuring of privacy policy documents by highlighting GDPR compliance and other legal concerns.

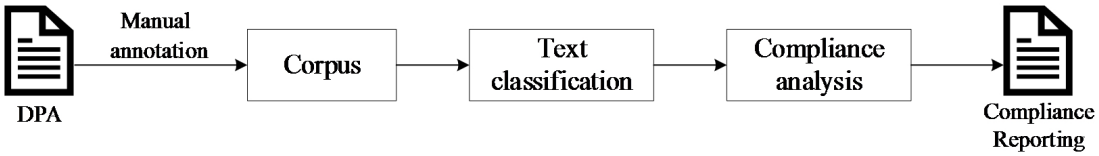

Figure 1: Overview of DPA compliance methods

3 DPA COMPLIANCE CHECKING

Although there have been some studies on privacy protection and data compliance, research on NLP technology specifically applied to GDPR compliance is still relatively scarce. This lack of GDPR specific NLP solutions reflects a broad research gap that urgently needs to be filled through innovative methods. Therefore, our research not only fills this gap, but also provides an advanced technical framework to automate and address the complex challenges of GDPR compliance.

The compliance detection process can be divided into two main steps, as shown in Figure 1. Firstly, the text classification task of NLP algorithm is used to manually label each sentence in DPA, assigning it a label related to GDPR, such as "personal data collection" or "purpose of personal data". Secondly, perform consistency detection based on GDPR rules, which analyzes the text in DPA according to GDPR regulations. For example, if there is a description about "personal data collection", the system will check if there is a corresponding user notification description. Through this method, we can evaluate whether the privacy protocol meets the requirements of GDPR.

We have adopted automatic web crawler technology to systematically collect privacy policy texts on the network. Specifically, through a web crawler program, the privacy policy page of the target website is automatically accessed, effectively extracting privacy policy text from hundreds of websites, and saving it in a local database for subsequent analysis. In the process of collecting privacy policies through web crawlers, the quality of some captured documents is not ideal, which affects the subsequent data analysis and processing effectiveness. To this end, a filtering phase was implemented with the aim of filtering out low-quality or irrelevant documents from the captured data, retaining only 200 privacy policies with complete content and clear structure. The filtered document will be used for the next detailed annotation process to ensure the accuracy and reliability of the dataset during the analysis process.

In the study of privacy policies, manual annotation is a crucial step that involves detailed identification and classification of key information in policy texts. In 2016, Wilson et al. hired multiple legal experts for manual annotation and created a public privacy policy corpus called OPP-115 [21]. The annotation is divided into ten categories, including "First Party Collection/Use", "Third Party Sharing/Collection", "User Choice/Control", etc. This classification method is also adopted in this article. We have organized a team of manual annotators who work based on a pre-defined set of detailed annotator guidelines that detail how to identify important privacy elements such as various data processing activities, user rights, and third-party sharing. By manually annotating these privacy policies, a high-quality and high-precision training dataset can be generated.

We used Long Short-Term Memory Network (LSTM) [10] and Bidirectional Encoder Representational Transformer (BERT) [22] models to evaluate and compare the effectiveness of these two advanced deep learning techniques in privacy policy text classification tasks. LSTM is suitable for processing and predicting important events in time series data, while BERT utilizes its powerful contextual recognition ability to understand and classify complex text data more accurately.

BERT is a revolutionary natural language processing model proposed by Google in 2018. It is based on the Transformer architecture and uses self-attention mechanism to process a large amount of unlabeled text data for pre training, in order to learn deep level features of language. The core innovation of BERT lies in its bidirectional training mechanism, which allows the model to simultaneously consider the left and right contexts of each word in the input text.

The pre training of BERT includes two main tasks: Masked Language Model (MLM) and Next Sentence Prediction (NSP). In MLM, the model randomly masks certain words in the input sentence and attempts to predict them. The formula can be expressed as:

$$L_{\text{MLM}} = -\frac{1}{N}\sum_{i=1}^{N}\log P(w_i|C_i) \tag{1}$$

where, N is the number of masked words, w_i is the i-th masked word, C_i is the context of the i-th masked word, and $P(w_i|C_i)$ is the probability distribution predicted by the model. In NSP tasks, the model learns to predict whether two sentences are continuous text relationships, which helps to understand the logical relationships between sentences. The loss function is defined as:

$$L_{\text{NSP}} = -\frac{1}{M}\sum_{j=1}^{M}\log P(y_j|S_j) \tag{2}$$

where, M is the number of sentence pairs, y_j is the true label of the j-th sentence pair (0 represents NotNext, 1 represents IsNext), S_j is the j-th sentence pair, and $P(y_j|S_j)$ is the probability distribution predicted by the model.

In multivariate text classification, text can only be classified into one category, and Softmax cross entropy is usually used as the loss function. In multi label text classification, a text can belong to multiple categories at the same time, and Sigmoid cross-entropy is used to process predictions for each category. If the total number of labels is significantly greater than the actual number of labels, it will lead to category imbalance. In this case, the loss of non-target categories may dilute the valuable loss of target categories, leading to a decrease in model training efficiency and slower convergence speed.

Table 1: Model Classification Results.

Label	LSTM			BERT		
	P	R	F1	P	R	F1
First Party Collection/Use	71.65	45.92	55.97	68.24	51.03	58.39
Third Party Sharing/Collection	75.43	41.49	53.53	74.51	69.83	72.09
User Choice/Control	62.75	37.86	47.23	63.62	47.83	54.61
User Access, Edit, & Deletion	80.45	77.27	78.83	74.75	70.16	72.38
Data Retention	70.15	63.39	66.60	76.18	69.53	72.70
Data Security	66.19	68.32	67.24	72.08	64.72	68.20
Policy Change	75.28	70.44	72.78	72.05	66.91	69.38
Do Not Track	61.93	59.30	60.59	64.35	68.71	66.46
International & Specific Audiences	53.76	41.04	46.55	58.91	44.32	50.58
Other	85.53	99.04	91.79	91.74	94.85	93.27

The definition of Softmax cross entropy is as follows:

$$loss = -\sum_{i=1}^{n} p(t) \log \frac{e^{s_t}}{\sum_{i=1}^{n} s_i} \tag{3}$$

where s is the output score, and p is the actual one hot label value for each position. Using Softmax cross-entropy deformation, starting from the essence of the cross-entropy formula, the above formula is only fixed at the meaningful position where $P(t)$=1. Softmax is transformed into the difference between the target predicted value and the predicted value at other positions, forming the form of LogSumExp, where St represents the predicted score of the target label as 1, and Si represents the predicted score of the other label as 0.

$$-\log \frac{e^{s_t}}{\sum_{i=1}^{n} s_i} = -\log(\frac{e^{s_t}}{\sum_{i=1}^{n} e^{s_i - s_t}}) = \log \sum_{i=1}^{n} e^{s_i - s_t} \tag{4}$$

In a single label scenario, it is expected that s_t will be larger than s_n at other positions, while in a multi-label scenario, all scores in the positive example set will be greater than each score in the negative example set corresponding to each positive example. The formula is as follows:

$$loss = \log(1 + \sum_{i \in \Omega_{neg}, j \in \Omega_{pos}} e^{s_i - s_j} + \sum_{i \in \Omega_{neg}} e^{s_i - s_0} + \sum_{j \in \Omega_{pos}} e^{s_0 - s_j}) \tag{5}$$

An additional s_0 score was introduced to represent a threshold, with the expectation that all negative case scores can be less than s_0 and all positive case scores can be greater than s_0. If a threshold of 0 is specified, it is simplified as:

$$loss = \log(1 + \sum_{i \in \Omega_{neg}} e^{s_i}) + \log(1 + \sum_{j \in \Omega_{pos}} e^{-s_j}) \tag{6}$$

this formula essentially compares the differences between the scores of all labeled categories and non-target categories, using the LogSumExp function to aggregate them into the maximum value, automatically eliminating the effects of non-maximal values, thus avoiding issues of class imbalance.

After effectively classifying privacy policy texts through NLP technology, we further designed a consistency detection method to evaluate whether these policies comply with GDPR regulations.

Firstly, we have defined a set of consistency rules based on the main requirements of GDPR. The main terms involved in these rules include transparency in data collection, clarity of data usage purposes, and clear acquisition of user consent. For example, we have set the following rule: if there is "First Party Collection/Use" (category label A) in a policy, it must be explicitly mentioned that "User Access, Edit, & Deletion" (category label B) and "Data Security" (category label C) are both mentioned.

Using NLP classification results, we divide the text into multiple paragraphs, each with corresponding GDPR related labels. Next, we developed an algorithm that traverses each annotated paragraph and checks whether all relevant consistency rules are met. This algorithm uses logical verification to confirm whether all specified conditions are met simultaneously, to determine whether the privacy policy is compliant.

4 Experimental Result

This experiment was conducted on Ubuntu LTS 22.04 operating system, with Intel Platinum 8336C CPU and GeForce RTX 4080 GPU. Implement LSTM and BERT neural network classification models using the Pytorch framework. We have configured a single-layer LSTM with 256 hidden units. The input of the LSTM model is obtained by embedding pre trained GloVe word vectors, with the word vector dimension set to 100. BERT uses the BERT based version, which includes 12 Transformer blocks to fine tune the BERT model to adapt to specific DPA classification tasks.

We selected a batch size of 32 for both training and validation. The initial learning rate for the models was set at 5e-5, with updates managed using the Adam optimizer. Both models were trained over 10 epochs, with evaluations of model performance on the validation set conducted at the end of each epoch to monitor training progress and prevent overfitting.

To ensure the fairness and effectiveness of model training and testing, the data in the corpus is randomly divided into training, validation, and testing sets, with proportions of 70%, 15%, and 15%, respectively. The training set is used for model training and parameter adjustment, the validation set is used for model selection and hyperparameter optimization, and the test set is used for the final evaluation of model performance. The main indicators for evaluating model performance include accuracy, recall, and F1 score.

Table 2: Compliance Testing Results.

	TP+FP	TP	P	R	F1
LSTM	612	526	85.95	87.00	86.47
BERT	601	531	88.35	90.50	89.41

These indicators will help us comprehensively evaluate the accuracy and efficiency of the model in identifying GDPR compliance requirements in DPA documents.

Table 1 shows the performance of LSTM and BERT in privacy policy text classification, and the results show that BERT outperforms LSTM in most classification tasks, especially in recall and F1 score. Although LSTM has slightly higher accuracy in certain categories, BERT is more balanced in overall performance.

According to the compliance analysis rules in Section 3, this section conducts compliance analysis on the text classification results. If there is "First Party Collection/Use" (category label A) in the policy, it is necessary to clearly mention "User Access, Edit, & Deletion" (category label B) and "Data Security" (category label C). The 200 privacy policies tested in this article are all from online services that are universally applicable to GDPR. Table 2 presents the results of compliance testing, including a performance comparison between LSTM and BERT models. This method successfully detected over 600 compliance issues in 200 privacy protocol files, with an accuracy of 85.95% for the LSTM model and 88.35% for the BERT model. The F1 score shows that the LSTM model is 86.47%, while the BERT model is 89.41%. It can be seen that the BERT model outperforms the LSTM model in all indicators and exhibits higher performance.

5 Conclusion

This study demonstrates the effectiveness of employing the BERT text classification model for conducting GDPR compliance assessments within DPA. Our method leverages a BERT-based model to evaluate GDPR compliance in data processing protocols, achieving an accuracy rate close to 86% and a recall rate exceeding 88%. These automated techniques significantly outperform traditional manual methods by accurately identifying and categorizing terms related to GDPR compliance. By automatically pinpointing key privacy elements, aligning them with GDPR mandates, and identifying discrepancies with high precision and recall, our approach considerably enhances the compliance assessment process. This research contributes to the rapidly advancing field of legal technology by offering a reliable and efficient solution for ensuring that data processing practices adhere to stringent privacy regulations, thus fostering improved data governance and legal compliance. Future efforts will focus on refining this methodology and expanding its application across various legal domains and document types.

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